Experiment-8

NMAP Live Host Discovery

AIM:

To discover live hosts using Nmap scans based on ARP, ICMP, TCP and UDP on the TryHackMe Nmap platform.

PROCEDURE:

1. Open the Nmap room on TryHackMe.
2. Study the target network and subnet structure.
3. Identify the target IP addresses.
4. Perform ARP-based host discovery.
5. Use:

   nmap -PR -Sn TARGET
6. Perform ICMP Echo discovery using:

   nmap -PE -sn TARGET
7. Perform ICMP Timestamp discovery

   nmap -PP -sn TARGET
8. Perform TCP SYN Ping using:

   nmap -PS23 TARGET
9. Perform TCP ACK Ping using the appropriate -PA option.
10. Study UDP-based host discovery.
11. Observe the live hosts identified by Nmap.

12. Study reverse DNS lookup using -R.

The manual covers ARP, ICMP, TCP and UDP live-host discovery and specifically identifies options such as -PR, -PE, -PP, -PM, -PS and -PA.

Sample Output:

```
Starting Nmap scan....
Nmap scan report for 10.10.210.5
Host is up.
Nmap scan report for 10.10.210.6
Host is up.
Nmap done:
2 hosts up
```

Result:

Thus, NMap successfully used to discover live hosts using ARP, ICMP, TCP and UDP - based Techniques.